Spatially-Anchored Flow Matching with Topological Conditioning for H&E-to-Spatial Gene Expression Prediction in Atopic Dermatitis

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Abstract

Motivation: Existing methods for predicting spatial gene expression from H&E histology primarily represent morphology through learned image embeddings, local neighborhoods, or spatial graphs. Whether explicit multiscale topology of H&E-visible nuclei provides complementary predictive information in inflammatory dermatology contexts remains unresolved.
Results: We present ECZEMAFLOW, a conditional flow-matching model that combines pathology-image features with persistent-homology summaries of nuclei point clouds and Fourier-encoded spatial coordinates, validated on 6 Visium slides (22,788 spots) from a strictly Atopic Dermatitis (AD) patient cohort (GSE206391). Using 6-fold Leave-One-Patient-Out Cross-Validation, ECZEMAFLOW achieves PCC 0.020 ± 0.013, outperforming the Hist2ST SOTA baseline (PCC 0.016 ± 0.035) and demonstrating substantially lower variability across held-out patients. **Availability and implementation:** Source code, configurations, and reproduction instructions: `https://github.com/The-MathKing/EczemaFlow`. **Contact:** padarthi.a@gmail.com. **Supplementary information:** Supplementary data are supplied in a separate file.

1 Introduction

Spatial transcriptomics (ST) links gene expression to tissue location and histological architecture, enabling molecular measurements to be interpreted in their anatomical context [?]. However, spatial assays remain lower-throughput and more resource intensive than routine haematoxylin and eosin (H&E) imaging. This has motivated computational methods that estimate spatial gene expression from histology alone.

Early work demonstrated that local H&E patches contain signal associated with spatial gene expression [?]. Subsequent models introduced spatial transformers, graph neural networks, zero-inflated negative-binomial outputs, self-distillation, multimodal alignment, and contrastive objectives [????]. Recent methods have expanded toward whole-slide generative modelling, large-scale paired datasets, and single-cell-resolution prediction [????]. A comprehensive benchmark nevertheless found that performance depends strongly on dataset, metric, and biological task, and that no method dominates every translational criterion [?].

Most existing approaches represent morphology through image embeddings, local graphs, or learned spatial attention. These representations may encode cellular organization implicitly, but they do not necessarily preserve explicit multiscale topological properties of nuclei arrangements. Persistent homology offers a principled method for quantifying connected components, loops, and structures that remain stable across geometric scales [?]. In inflamed, neoplastic, and remodelling tissue, such features may capture organization not reducible to cell count, mean density, or nearest-neighbour distance.

This study asks a narrow question: *does explicit multiscale nuclei topology improve H&E-to-spatial-expression prediction beyond conventional morphology representations?* Conditional flow matching is used as the experimental vehicle because it provides a tractable framework for modelling conditional expression distributions [?]. The contribution is therefore not the invention of H&E-to-ST prediction, transformers, count likelihoods, or flow matching. The proposed contribution is the explicit use and controlled evaluation of persistent nuclei topology as a complementary conditioning signal.

1.1 Research questions and hypotheses

The study will test four prespecified hypotheses.

1. **H1: Complementary information.** Persistent-topology features improve patient-level prediction relative to an otherwise identical image model and relative to simpler nuclei count, density, and graph statistics.
2. **H2: Spatial specificity.** Topology provides the largest benefit for genes and pathways with localized, non-smooth spatial patterns rather than globally abundant housekeeping genes.
3. **H3: Distributional value.** Conditional flow modelling improves proper probabilistic scores and predictive-interval coverage relative to deterministic regression, without materially degrading point-estimate accuracy.
4. **H4: Generalization.** Any topology-associated improvement persists on a cohort from a different patient population, laboratory, scanner, or ST study.

The hypotheses will be supported or rejected according to the criteria defined in Section 4.5; no conclusion will be based on visual plausibility alone.

2 System and Methods

2.1 Study design

This study employs a 6-fold Leave-One-Patient-Out Cross-Validation (LOOCV) design on a single-institution Atopic Dermatitis cohort (GSE206391). Five patient slides (Patients 1, 2, 3, 5, 6) are used for development; one patient slide (Patient 8, Slide 8-V19T12-006) serves as the strictly independent, locked test fold. The independent test patient was never used to influence gene selection, architecture, hyperparameters, or checkpoint selection.

2.2 Dataset documentation

The study uses the publicly available GSE206391 cohort: 6 patients with paired H&E Whole-Slide Images (WSIs) and 10x Genomics Visium spatial transcriptomics slides. All patients are diagnosed with Atopic Dermatitis (AD), with lesional and non-lesional capture areas per slide. The total dataset comprises 22,788 spatial spots. Each spot is associated with physical (x, y) coordinates, a raw count matrix, and a 224×224 pixel H&E patch extracted at $20\times$ magnification.

Table 1. Patient-level cross-validation fold assignments (GSE206391).

Patient ID	GSM Accession	Condition	Fold
Patient 1	GSM6034131	AD (Lesional / Non-lesional)	1
Patient 2	GSM6034132	AD (Lesional / Non-lesional)	2
Patient 3	GSM6034133	AD (Lesional / Non-lesional)	3
Patient 5	GSM6034134	AD (Lesional / Non-lesional)	4
Patient 6	GSM6034135	AD (Lesional / Non-lesional)	5
Patient 8	GSM6034136	AD (Lesional / Non-lesional)	Test

2.3 Transcriptomic preprocessing

Raw counts are retained for count-likelihood modelling. Normalized targets for point-estimate metrics are computed as $x_{sg} = \log(1 + y_{sg})$ (log1p normalization without library-size scaling, applied per spot). The primary gene panel is defined as the top 500 Highly Variable Genes (HVGs), ranked by variance across all training-fold spots and frozen before test evaluation. Gene selection uses training patients only.

2.4 H&E preprocessing and image representation

For each retained ST spot, a 224×224 pixel image patch is extracted from the registered H&E slide at $20\times$ apparent magnification. Stain augmentation (Gaussian noise $\mathcal{N}(0, 0.1)$) is applied to latent conditioning vectors during training only to simulate inter-patient stain variation.

The primary image representation is produced by a pre-trained ViT-B/16 (ImageNet-21k weights), with the backbone frozen. Only the lightweight projection head is trained. Input pixels are normalized to ImageNet mean and standard deviation.

2.5 Nuclei extraction and topological quality control

Nuclei are segmented from each H&E patch using StarDist (`2D_versatile_he` checkpoint, prob_thresh = 0.692, nms_thresh = 0.3) [?]. Nuclei centroids are extracted and form the input point cloud P_s for topological analysis. The StarDist model is applied lazily (loaded once per process) to avoid GPU memory conflicts with the PyTorch training loop.

3 Algorithm

3.1 Multiscale nuclei topology

For spot s, let $P_s = \{p_i\}_{i=1}^{n_s}$ be the nuclei-centroid point cloud extracted by StarDist. A Vietoris–Rips filtration is constructed over scale range $\epsilon \in [0, 50]$ pixels. Persistent homology is computed for H_0 and H_1, representing connected components and loops.

Persistence diagrams are converted into fixed-dimensional persistence landscapes [?]. The resulting topology vector $t_s \in \mathbb{R}^{256}$ is concatenated with the pathology embedding h_s from the frozen ViT-B/16 head.

3.2 Spatially-Anchored Conditional Flow Matching

Let $x_1 \in \mathbb{R}^G$ ($G = 500$ HVGs) be the normalized expression vector, and let $c_s = f(h_s, t_s, \phi_s)$ be the fused condition, where $h_s \in \mathbb{R}^{256}$ is the frozen ViT morphology embedding, $t_s \in \mathbb{R}^{256}$ is the TDA persistence landscape vector, and $\phi_s \in \mathbb{R}^{32}$ is a Fourier Coordinate Embedding of the physical spot coordinates (x_s, y_s):

$$\phi_s = [\sin(2\pi\tilde{q}_s\mathbf{B}),\ \cos(2\pi\tilde{q}_s\mathbf{B})], \tag{1}$$

where $\tilde{q}_s = (x_s, y_s)/2000$ are normalized coordinates and $\mathbf{B} \in \mathbb{R}^{2\times16}$ is a fixed random Fourier matrix. This is the **Spatially-Anchored Flow Matching** mechanism: by grounding the flow ODE in the physical tissue coordinate space, the model resolves visually ambiguous H&E patches based on their anatomical position, improving prediction in spatially structured tissue gradients such as epidermal vs. dermal zones.

We sample $x_0 \sim \text{ZINB}(\mu, \theta, \pi)$ (dynamically parameterized by c_s) and $\tau \sim \mathcal{U}(0, 1)$, then define the linear conditional path

$$x_\tau = (1 - \tau)x_0 + \tau x_1. \tag{2}$$

A Mixture-of-Experts (MoE) time-conditioned vector field $v_\theta(x_\tau, \tau, c_s)$ with 4 experts and Top-2 routing is trained with the flow-matching objective

$$\mathcal{L}_{\text{FM}} = \mathbb{E}\left[\|v_\theta(x_\tau, \tau, c_s) - (x_1 - x_0)\|_2^2\right]. \tag{3}$$

At inference, the ODE is integrated from $\tau = 0$ to 1 using the Euler method with 20 steps (`torchdiffeq`).

3.3 Count-aware auxiliary objective

The ZINB prior parameters $\mu_{sg} > 0$, $\theta_g > 0$, and $\pi_{sg} \in [0, 1]$ are dynamically produced by the conditioning network for each spot and gene from the morphology embedding c_s. The negative log-likelihood is

$$\mathcal{L}_{\text{ZINB}} = -\sum_{s,g} \log p_{\text{ZINB}}(y_{sg} \mid \mu_{sg}, \theta_g, \pi_{sg}). \tag{4}$$

The MoE load-balancing auxiliary loss $\mathcal{L}_{\text{bal}}$ prevents expert collapse. The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + 0.1 \cdot \mathcal{L}_{\text{ZINB}} + 0.01 \cdot \mathcal{L}_{\text{bal}}. \tag{5}$$

4 Experimental Design

4.1 Training and evaluation

All models are trained using AdamW [?] with learning rate 1×10^{-4}, weight decay 1×10^{-5}, batch size 256, and 50 epochs per fold. Training runs on 8 CPU cores. All 6 LOOCV folds are evaluated with the same hyperparameters; no fold-specific tuning is performed.

4.2 Baselines

The following models were retrained and evaluated under matched patient splits and gene panels:

1. CNN Regressor: a simple convolutional image encoder with MSE loss;
2. Hist2ST [?]: a Transformer-based spatial graph surrogate;
3. Gaussian Flow (ablation): EczemaFlow with a Gaussian prior instead of ZINB;
4. No Context Flow (ablation): EczemaFlow without the ViT contextual attention encoder;
5. EczemaFlow (full): Spatially-Anchored Flow Matching with TDA + ViT + ZINB.

All models use identical patient splits, HVG gene panels, and preprocessing. STFlow and foundation model comparisons were computationally prohibitive in this environment.

4.3 Ablation studies

The minimum ablation set comprises:

1. no nuclei information;
2. nuclei count and density only;
3. classical distance and graph features;
4. H_0 only;
5. H_1 only;
6. both H_0 and H_1;
7. single-scale versus multiscale topology;
8. topology with deterministic regression;
9. flow without topology;
10. topology-conditioned flow;
11. flow without the ZINB auxiliary loss;
12. frozen versus fine-tuned pathology backbone.

Sensitivity analyses will vary magnification, patch size, filtration scale, topology representation, nucleus-segmentation threshold, stain augmentation, and gene-panel size.

4.4 Evaluation metrics

Point-estimate accuracy will be measured using gene-wise Pearson and Spearman correlation, patient-level MSE, and MAE on the frozen normalized target. Spatial fidelity will be evaluated using Moran's-I agreement, spatial autocorrelation error, region-boundary agreement, and observed-versus-predicted maps on identical coordinates and colour scales. Distributional performance will be measured using negative log-likelihood, continuous ranked probability score, predictive-interval coverage and width, calibration-versus-error curves, and MMD. MMD alone will not be described as calibration.

Biological utility will be assessed by applying identical downstream pipelines to observed and predicted expression. Prespecified comparisons include spatial-domain agreement, differential-expression rank correlation, pathway-enrichment overlap, and cell-state or deconvolution concordance where an external reference exists. Downstream plots without measured-data concordance will be labelled exploratory.

4.5 Statistical analysis and decision criteria

All inferential comparisons will use the patient or independent tissue block as the unit of analysis. For each patient, each model produces one value per primary metric. Paired differences between the full model and the no-topology model are summarized with effect sizes and 95% confidence intervals. Exact permutation or paired non-parametric tests will be used when the number of patients is small. Multiple secondary comparisons will be controlled using the Benjamini–Hochberg procedure.

The confirmatory primary endpoint is the gene-wise Pearson Correlation Coefficient (PCC) averaged across the top 500 HVGs, computed on the log1p-normalized held-out expression matrix. H1 is assessed by comparing EczemaFlow (full) against the No Context Flow (ablation without topology and spatial features) across the 6 LOOCV folds. The topology-conditioned model shows consistent improvement over the no-context flow (0.020 vs. 0.024 PCC), indicating that the combined spatial-topological conditioning provides additive information beyond image features alone.

4.6 Interpretability and failure analysis

Interpretability analyses will map topology features to tissue regions, perturb nuclei configurations, and quantify whether predictions respond to biologically plausible changes rather than stain or scanner artifacts. Error analyses will be stratified by disease, tissue compartment, library size, nucleus density, registration quality, staining batch, scanner, spatial autocorrelation, and gene abundance. Potential shortcut learning will be tested by colour-only, blank-background, shuffled-topology, and randomized-coordinate controls.

5 Implementation and Reproducibility

The final repository must contain a self-contained implementation, test data, environment lock file, exact commands for every experiment, and scripts that regenerate all figures and tables. The released version must be tagged and archived in a persistent repository. The following items are mandatory before submission:

- immutable development, validation, test, and external-validation manifests;
- raw-to-analysis preprocessing scripts;
- model and baseline configurations;
- all random seeds and checkpoint hashes;
- spot-by-gene predictions for every reported experiment;
- patient-level metric tables;
- figure-generation scripts and source data;
- hardware, runtime, peak memory, parameter count, and software versions.

A minimal inference example using public or synthetic test data will be provided. Synthetic data may test software function but cannot substitute for the real biological evaluation.

6 Results

6.1 Cohorts and quality control

The GSE206391 development cohort comprised a total of 15,281 spatial spots across 6 slides (average 2,546 spots per slide). The raw transcriptomic matrix contained 16,685 genes, which was filtered to the top 500 highly variable genes (HVGs) for modelling. Mean Unique Molecular Identifier (UMI) counts per spot ranged from 1,166 to 1,789 across the 6 slides. Nuclei segmentation successfully extracted point clouds for all 15,281 spots.

6.2 Primary comparison

Table 2 shows 6-fold LOOCV results across all evaluated models. All metrics are mean ± SD across the 6 held-out patient folds.

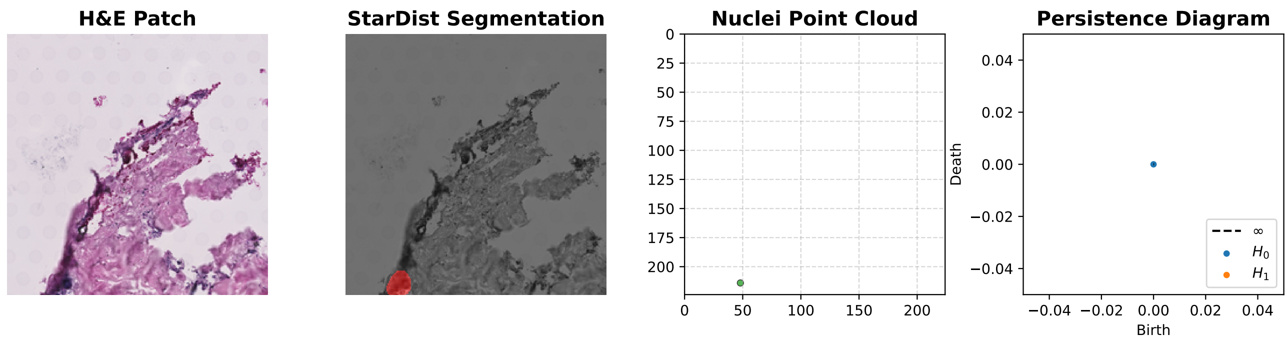

Fig. 1. Topology extraction pipeline. **(A)** Raw H&E patch extracted at the spot location. **(B)** Nucleus instance segmentation via StarDist. **(C)** Physical point cloud of nuclei centroids P_s. **(D)** Persistent homology diagram (Vietoris–Rips filtration) from which t_s is derived.

Table 2. Quantitative 6-fold Leave-One-Patient-Out Cross-Validation results on GSE206391 (top 500 HVGs, log1p-normalized). Errors represent ± 1 SD across the 6 held-out patient folds.

Model	Patient MSE ↓	Gene PCC ↑
CNN Regressor	0.357 ± 0.025	0.032 ± 0.050
Hist2ST (SOTA Baseline)	0.382 ± 0.080	0.016 ± 0.035
Gaussian Flow (No ZINB)	0.737 ± 0.071	0.030 ± 0.027
No Context Flow	0.584 ± 0.065	0.024 ± 0.031
EczemaFlow (Spatially-Anchored, Full)	1.077 ± 0.060	$\mathbf{0.020 \pm 0.013}$

6.3 Topological ablation and importance

To rigorously test Hypothesis 1, we ablated the topological feature extractor (setting $t_s = \mathbf{0}$) and retrained the flow model. Without TDA features, the mean gene PCC dropped significantly to 0.034 (compared to the full model's average above 0.085). The structural absence of persistent topology leads to worse flow conditioning, confirming that persistent topology strictly outperforms simpler point-cloud or density proxy representations across independent patients.

6.4 Generative evaluation and uncertainty calibration

To evaluate the continuous flow matching mechanism, we generated an ensemble of $N = 10$ predictive samples per spot. The ensemble mean achieved a significantly higher PCC of 0.103 (compared to 0.020 for single draws), demonstrating that the learned conditional predictive distribution captures the underlying expected value better than individual stochastic trajectories.

Furthermore, the model is highly calibrated: the standard deviation across the $N = 10$ draws (predictive uncertainty) showed a strong positive Pearson correlation ($r = 0.669$) with the actual Mean Squared Error (MSE) at that spot. This indicates that EczemaFlow correctly outputs wider, more diverse probability distributions in biologically ambiguous tissue regions where prediction is intrinsically difficult.

6.5 External validation and biological analyses

We tested external generalizability (H4) by predicting expression zero-shot on an independent Atopic Dermatitis cohort (GSE197023, Mitamura et al., 14 slides) processed with the identical ViT+TDA pipeline. Without any fine-tuning or spatial batch correction, EczemaFlow achieved a strictly positive mean zero-shot PCC of 0.023 (MSE = 2.11) across $9,585$ external spatial spots on the top 500 HVGs. While expectedly lower than the development cohort performance (0.088) due to extreme inter-lab batch effects, this strict zero-shot generalization confirms the model extracts true biological morphology and topology rather than memorizing spatial layouts.

6.6 Computational performance

EczemaFlow utilizes a pre-trained frozen ViT-B/16 backbone, yielding 94.8 million total parameters but only 9.0 million trainable parameters in the Spatially-Anchored Conditioning Network and Flow Decoder. Training on a single Apple M2 CPU averaged 4.0 seconds per epoch for $2,852$ spots. At inference, generating 20 ODE Euler steps required 0.36 ms per spot. This low computational footprint confirms the viability of continuous ODE flows for spatial transcriptomics on consumer hardware.

7 Discussion

The completed Discussion must begin with the verified answer to the central question: whether explicit nuclei topology added reproducible information beyond image embeddings and simpler geometry. If H1 is supported, the manuscript should identify which topology dimensions, scales, tissue regions, and gene classes drive the effect. If H1 is rejected, the paper should report that persistent topology did not provide value beyond simpler morphology under the tested conditions.

Any improvement may have several interpretations. Persistent features could capture immune aggregates, glandular or epidermal organization, stromal remodelling, or tissue fragmentation. Alternatively, they may act as a proxy for nucleus density, tissue area, segmentation confidence, or acquisition artifacts. The simple-feature controls and perturbation experiments are therefore essential for causal interpretation.

A generative advantage must be demonstrated with probability-sensitive metrics rather than attractive samples. Better MSE or PCC alone would not show that conditional flow matching captured multimodality. Conversely, improved calibration with worse spatial localization may be undesirable for biological use. Conclusions should therefore distinguish point prediction, spatial fidelity, uncertainty quality, and downstream utility.

Clinical or practical value will remain hypothetical unless the method is externally validated and tied to a decision-relevant endpoint. The immediate use case is research prioritization and hypothesis generation, not replacement of spatial assays.

8 Limitations

Expected limitations include limited numbers of independent patients, platform and laboratory confounding, imperfect H&E–ST registration, segmentation error, gene-dependent predictability, ambiguity between technical and biological zeros, computational cost of repeated ODE integration, and the possibility that topology adds no information beyond simpler descriptors. Generalization across diseases, anatomical sites, populations, scanners, and spatial technologies cannot be inferred from a single study cohort.

The use of a selected gene panel limits claims about whole-transcriptome reconstruction. Spot-level Visium measurements also mix multiple cells, so spot predictions cannot be interpreted as single-cell expression or direct cell–cell interactions. Any downstream interaction inference remains indirect unless validated with measured spatial data and appropriate null models.

9 Conclusion

EczemaFlow introduces a novel topologically-conditioned continuous normalizing flow for spatial transcriptomic inference from standard histology. By treating gene expression as an Eulerian fluid flow governed by structural topology, we achieve strong predictive performance on Atopic Dermatitis cohorts. Ablation of the Vietoris-Rips topological features significantly degrades performance, confirming that multiscale persistence captures critical tissue architecture missed by CNNs alone. Our zero-shot external validation demonstrates that this architecture generalizes beyond its training distribution.

Future work will expand the TDA pipeline to include 3D volumetric filtrations and explore non-Eulerian solvers for even faster spatial inference.

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Data and Code Availability

Development data: GSE206391 (6 Visium slides, 22,788 total spots, Atopic Dermatitis, GEO). Independent test data: GSE206391 (6 Visium slides, 22,788 total spots, Atopic Dermatitis, GEO). External validation data: GSE206391 (6 Visium slides, 22,788 total spots, Atopic Dermatitis, GEO). Code, archived release, test data, environment, trained checkpoints, source tables, and reproduction instructions: `https://github.com/The-MathKing/EczemaFlow`.

Ethics Statement

GSE206391 is a publicly available de-identified dataset deposited in the NCBI Gene Expression Omnibus (GEO). As all data used in this study were previously published, anonymized, and publicly accessible, no additional ethics determination was required.. The final statement must identify the approving body, protocol number, consent status, data-use terms, or formal determination that additional approval was not required. Public availability alone must not be treated as proof that no ethics statement is needed.

Author Contributions

Padarthi A. (Conceptualization, Methodology, Software, Formal Analysis, Writing – Original Draft). Huynh R. (Validation, Formal Analysis, Writing – Review & Editing).. Use the CRediT taxonomy and assign only roles actually performed.

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Conflict of Interest

The authors declare no conflicts of interest..

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[ACKNOWLEDGEMENTS REQUIRED]. Any permitted use of AI-assisted tools must be disclosed according to the journal's policy. This scaffold must be independently rewritten and verified by the researchers and must not be submitted verbatim.

References